

```
sum.py -- ~/hw.py

Users > rohitchevan > hw.py

Project
> sum.py
> External Libraries
> Scratches and Consoles

1 class Book:
2     def __init__(self, title, author):
3         self.title = title
4         self.author = author
5         self.available = True
6 class Patron:
7     def __init__(self, name):
8         self.name = name
9         self.borrowed_books = []
10 class Library:
11     def __init__(self):
12         self.books = []
13         self.patrons = []
14
15     def add_book(self, title, author):
16         self.books.append(Book(title, author))
17         print("Book added successfully!")
18
19     def register_patron(self, name):
20         self.patrons.append(Patron(name))
21         print("Patron registered successfully!")
22
23     def issue_book(self, title, patron_name):
24         for book in self.books:
25             if book.title == title and book.available:
26                 for patron in self.patrons:
27                     if patron.name == patron_name:
28                         book.available = False
29                         patron.borrowed_books.append(book.title)
30
31 while True:
32     else
```

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28         book.available = False
29         patron.borrowed_books.append(book.title)
30         print("Book issued successfully!")
31         return
32     print("Book unavailable or patron not found.")
33
34     def return_book(self, title, patron_name):
35         for book in self.books:
36             if book.title == title:
37                 for patron in self.patrons:
38                     if patron.name == patron_name and title in patron.borrowed_books:
39                         book.available = True
40                         patron.borrowed_books.remove(title)
41                         print("Book returned successfully!")
42                         return
43         print("Invalid return request.")
44
45     def display_books(self):
46         print("\nBooks:")
47         for book in self.books:
48             status = "Available" if book.available else "Issued"
49             print(f"{book.title} by {book.author} - {status}")
50
51     def display_patrons(self):
52         print("\nPatrons:")
53         for patron in self.patrons:
54             print(f"{patron.name} -> Borrowed Books: {patron.borrowed_books}")
55
56 while True:
57     else
```

The screenshot shows a PyCharm IDE window titled "sum.py - ~/hw.py". The left sidebar displays the project structure with "sum.py" selected. The main editor area shows the following Python code:

```
57 library = Library()
58
59 while True:
60     print("\n--- Library Management System ---")
61     print("1. Add Book")
62     print("2. Register Patron")
63     print("3. Issue Book")
64     print("4. Return Book")
65     print("5. Display Books")
66     print("6. Display Patrons")
67     print("7. Exit")
68
69     choice = input("Enter your choice: ")
70
71     if choice == "1":
72         title = input("Enter book title: ")
73         author = input("Enter author name: ")
74         library.add_book(title, author)
75
76     elif choice == "2":
77         name = input("Enter patron name: ")
78         library.register_patron(name)
79
80     elif choice == "3":
81         title = input("Enter book title: ")
82         name = input("Enter patron name: ")
83         library.issue_book(title, name)
84
85     elif choice == "4":
86         title = input("Enter book title: ")
87         name = input("Enter patron name: ")
88         library.return_book(title, name)
89
90     elif choice == "5":
91         library.display_books()
92
93     elif choice == "6":
94         library.display_patrons()
95
96     elif choice == "7":
97         break
98
99 if __name__ == "__main__":
100     library = Library()
101     main()
```

```
84
85 elif choice == "4":
86     title = input("Enter book title: ")
87     name = input("Enter patron name: ")
88     library.return_book(title, name)
89
90 elif choice == "5":
91     library.display_books()
92
93 elif choice == "6":
94     library.display_patrons()
95
96 elif choice == "7":
97     print("Exiting...")
98     break
99
100 else:
101     print("Invalid choice!")
```

```
/usr/local/bin/python3.9 /Users/rohitchavan/hw.py
```

```
--- Library Management System ---
```

1. Add Book
2. Register Patron
3. Issue Book
4. Return Book
5. Display Books
6. Display Patrons
7. Exit

```
Enter your choice: 1
```

```
Enter book title: Python Programming
```

```
Enter author name: John Smith
```

```
Book added successfully!
```